

Admin:

Final exam:

- 18 questions divided into three parts: A, B, C.
  - Each part has 6 questions and 60 min.
  - Submit 5 questions and exclude one.
  - Each question is worth 20 points.

Example submission:

PART A

|           |    |
|-----------|----|
| <u>AL</u> | ID |
|-----------|----|

|    |
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| A2 |
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| A3 |
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| A4 |
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|    |
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| A5 |
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|    |
|----|
| A6 |
|----|

Total grade:  $(A, B, C \times 5) \times 20 \text{pts} = 300 \text{ points}$

60 min working on exam, 5 min to upload (not supposed to work)

Repeat. 5 min break

BBB - individual conferences.

Share screen.

I'm uploading the exam in BBB

## Lecture 13:

## Quizizz

Brief review on bifurcation diagrams  
phase portraits sketch.

General questions

Q12

$$ay' + y = b$$

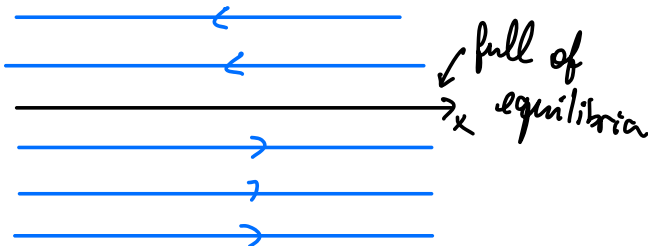
$$y' = \frac{b-y}{a} = 0 \Rightarrow y = b.$$

$$f(y) \quad f'(y) = -\frac{1}{a} < 0 \text{ sink.}$$

by the derivative  
criterion.

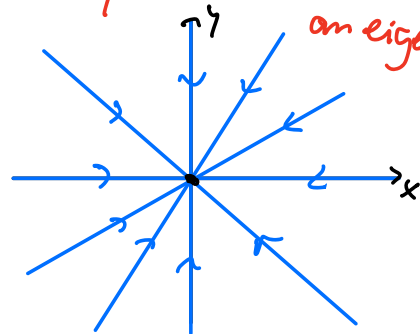


Q22  $x' = \begin{bmatrix} a & b \\ c & d \end{bmatrix} x$



$$x' = \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix} x$$

any non-zero vector is  
an eigenvector



$$A = -2 \cdot I$$

$$Av = -2I \cdot v = -2v.$$

Q15  $m, b, k \geq 0$   
 $m > 0$

spiral  
 sink  $\rightarrow$  overdamped / crit. damp.  
 underdamped.  
 source  
 saddle  
 center  $\rightarrow$  undamped

MATLAB3: All solutions for damped go to zero.  
 Undamped are periodic.

Charact. equation:  $m\lambda^2 + b\lambda + k = 0$   

$$\lambda = \frac{-b}{2m} \pm \frac{1}{2m} \sqrt{b^2 - 4km}$$

$$\leq \frac{b}{2m}$$

$\text{Re}(\lambda) < 0 \quad \underline{b > 0.}$

(20) d) is a solution for 2nd order linear with  
 constant coeffs. (not a "vector")

(18) degenerate source (rep. eigenvalues)

(16) 
$$\begin{cases} x' = y \\ y' = \frac{-cx - by}{a} \end{cases}$$

(0,0) is always a solution

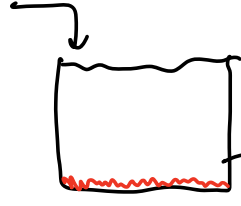
$$A = \begin{bmatrix} 0 & 1 \\ -c/a & -b/a \end{bmatrix}$$
 if A is invertible,  
 then  $\exists!$  solution

if A is singular, there are  
 infinite solutions.

(18)  $V(t) = 20 \cdot 5 \cdot 10 = 1000$   
 $C(0) = \frac{3}{100} \cdot 1000 = 30.$

$$\frac{dC}{dt} = 1 - \frac{C}{10}$$

$100 \text{ ft}^3 \cdot 1\% = 1$



$$\frac{dC}{dt} = \frac{1}{\text{in}} - \frac{C}{10} \text{ out}$$

(9) Integrating Factor:  $a(t) \cdot y' + b(t)y + c(t) = 0.$

LINEAR

Undetermined Coeffs: constant coeffs  $a, b.$   
 $ay' + by + c(t) = 0 \Rightarrow y' = -\frac{b}{a}y - \frac{c(t)}{a}$   
 $y_h(t) = D \cdot e^{\frac{1}{a}t}$

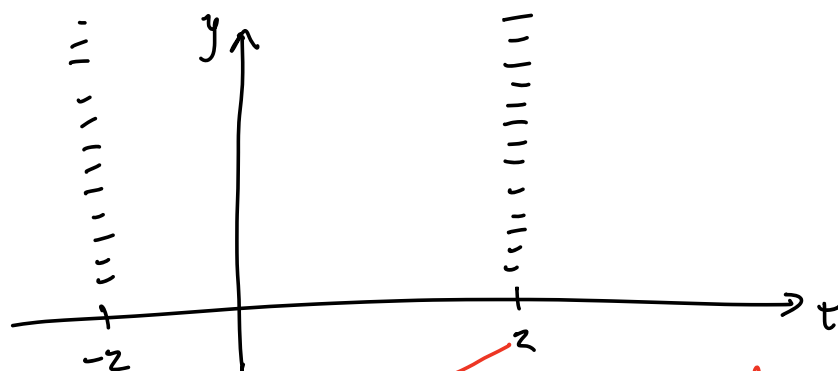
Separation of Variables:  $y' = f(y)g(t)$

(2) if  $t_0$  is such that

$$y' = 0 \Rightarrow t_0 \text{ is a critical point (Calc I).}$$

if  $y_0$  is such that

$$y' = 0 \Rightarrow y_0 \text{ is an equilibrium solution}$$

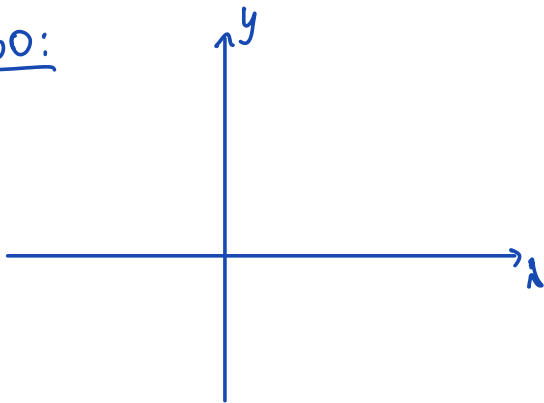


critical points = horizontal slope.

# Bifurcation Diagrams.

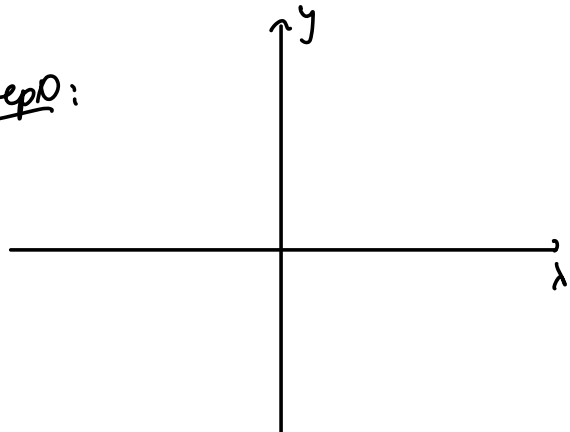
$$y' = -y^3 + (\lambda - 7)y$$

Step 0:



$$y' = f(\lambda, y), \lambda \in \mathbb{R}$$

Step 0:



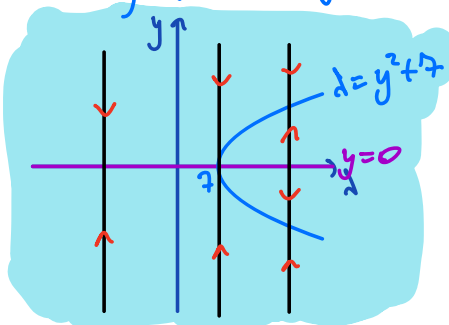
Step 1:

$$y' = 0 \Rightarrow -y^3 + (\lambda - 7)y = 0$$

$$y(-y^2 + (\lambda - 7)) = 0$$

$$y = 0 \quad \text{or} \quad y^2 = \lambda - 7$$

$$y = \pm \sqrt{\lambda - 7} \quad \lambda = y^2 + 7$$

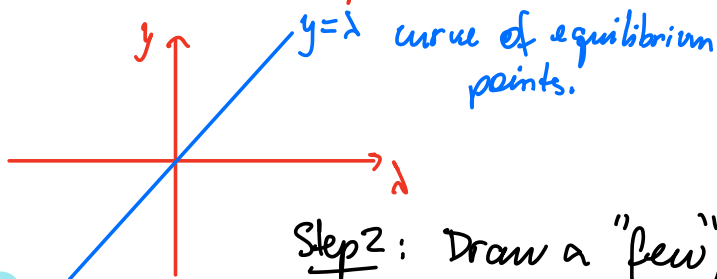


Parenthesis

$$y' + y = \lambda$$

$y' = 0$  and find equilibria

Step 1: Draw the curves of equilibria.



Step 2: Draw a "few" phase lines

$$f(y, \lambda) = -y^3 + (\lambda - 7)y$$

$$\frac{\partial f}{\partial y} = -3y^2 + (\lambda - 7)$$

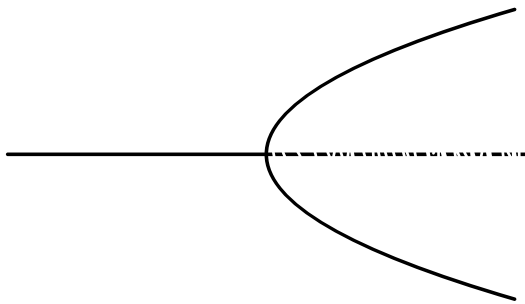
$$\lambda = 0$$

$$f(y) = -y^3 - 7y \Rightarrow f'(y) = -3y^2 - 7$$

$$y = 0 \Rightarrow f'(0) = -7 \text{ sink.}$$

$$\lambda = 10$$

$$f'(10) = \lambda - 7 = 10 - 7 = 3 \Rightarrow \text{source at } y = 0.$$



pitchfork bifurcation.

\* end of bif. diagrams.

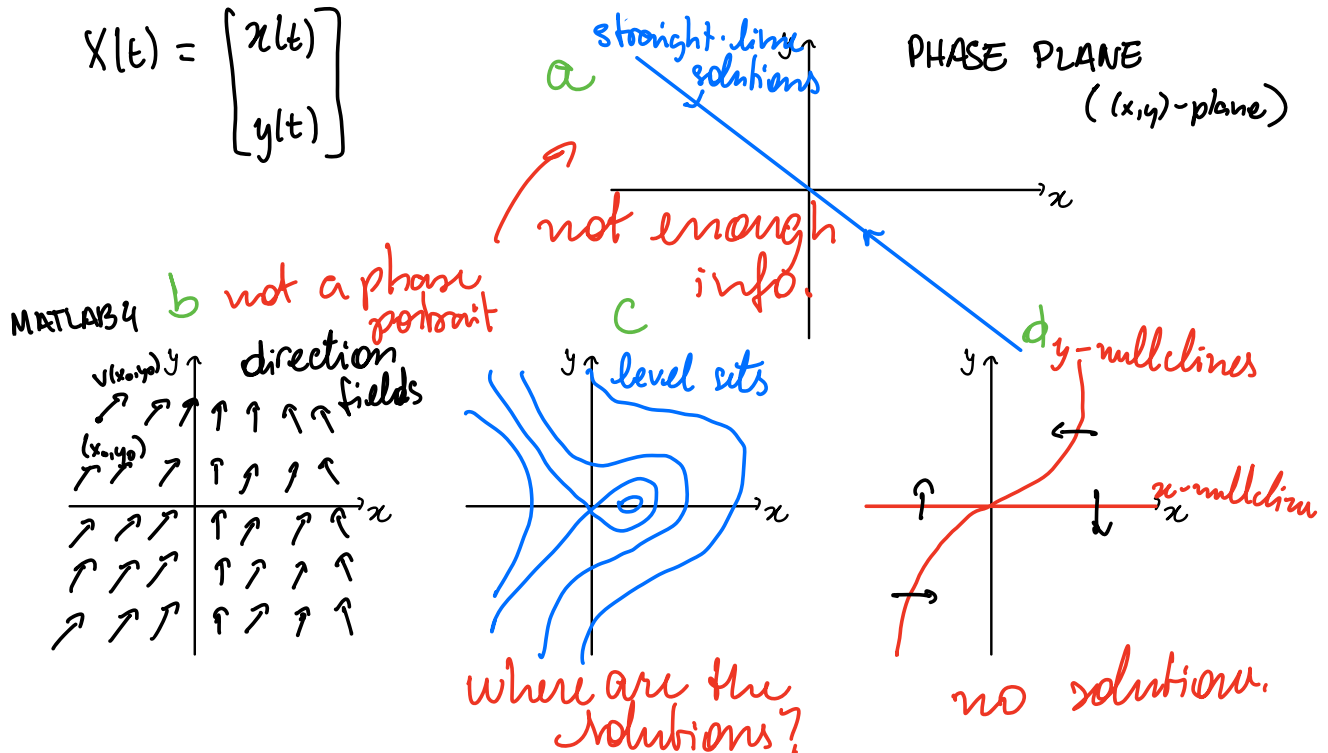
break: 7:57.

## Recap: Phase Portrait.

Motivation: Geometry of solutions give more information than the analytical solution.

$$X(t) = c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} p \\ 1 \end{bmatrix} e^{3t}$$

$$X(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$$



Ex Draw the phase portrait of *where do they go?*

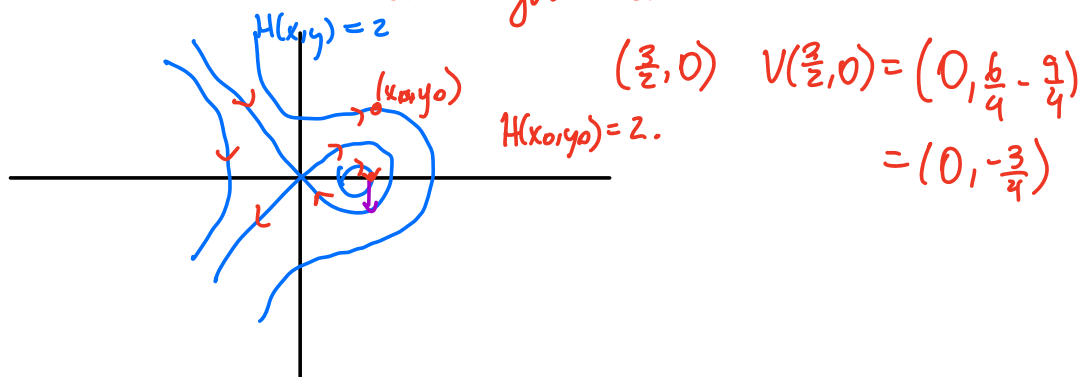
$$x' = y$$

$$y' = x - x^2$$

phase portrait: sample of solutions/trajectories in phase plane.

- curve
- arrows

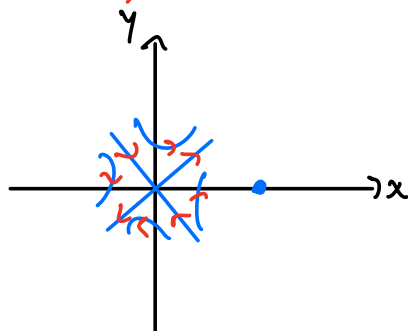
- Strategy 1: If you find  $H(x,y)$  s.t. solutions are contained in level sets, then draw level sets and give direction.



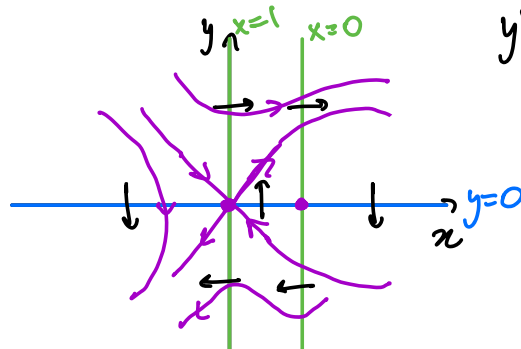
- Strategy 2: Check linearization of equilibrium points, draw it locally and hope for the best.

$(0,0)$  is a saddle,

$(1,0)$  doesn't give any results.



• Strategy 3: nullclines



$$\begin{aligned} x' &= y \Rightarrow y=0 \\ y' &= x - x^2 \Rightarrow x^2 - x = 0 \\ x(1-x) &= 0 \\ x &= 0, x=1. \end{aligned}$$

-end of phase portrait  $x' = Ax, X(t_0) = X_0$

$$A = PJP^{-1}$$

$$\begin{aligned} e^{At} X_0 &= \begin{cases} c_1 v_1 e^{\lambda_1 t} + c_2 v_2 e^{\lambda_2 t} & \lambda_1 \neq \lambda_2 \\ c_1 v e^{t} + c_2 w t e^{t} & \lambda \text{ repeated} \\ c_1 \operatorname{Re}(v e^{\lambda t}) + c_2 \operatorname{Im}(v e^{\lambda t}) & \lambda, \bar{\lambda} \text{ complex} \end{cases} \\ P e^{Jt} P^{-1} X_0 &= \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \end{aligned}$$

don't forget that  $A^{-1} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \cdot \frac{1}{ad-bc}$   
when  $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ .

$\lambda_1 \neq \lambda_2$

$$\begin{bmatrix} v_1 & v_2 \end{bmatrix} \begin{bmatrix} e^{\lambda_1 t} \\ e^{\lambda_2 t} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = c_1 v_1 e^{\lambda_1 t} + c_2 v_2 e^{\lambda_2 t}.$$

Reminder: OFFICE HOURS :

Thursday 4:40-5:40pm

Friday: 10:30-11:30am



Lyapunov ~ Gradient.

- $L(x,y)$  is Lyapunov for the system

$$\begin{cases} x' = f(x,y) \\ y' = g(x,y) \end{cases}$$

if it is decreasing along solutions.

- A Gradient System is

$$\begin{cases} x' = \frac{\partial G}{\partial x} \\ y' = \frac{\partial G}{\partial y} \end{cases}$$

for some  $G(x,y)$ .

$G$  is increasing along solutions  $\Rightarrow L = -G$  is a Lyapunov function

for  $\begin{cases} x' = G_x \\ y' = G_y \end{cases}$

